MedRetGen: Retrieval-Augmented Generation for Medical Question Answering

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Abstract

MedRetGen applies the RetGen retrieve-then-generate pipeline to Medical QA. DenseRet indexes medical textbooks and guidelines, retrieves passages for each question, and supplies them to CuraLM-70B as evidence. The approach improves CuraLM-70B by about four accuracy points on MedQA and also helps on PubMedQA. Retrieval makes relevant facts available at inference time, but passages sometimes contradict one another and the generator cannot always reason across multiple facts. MedRetGen therefore improves single-fact questions more reliably than synthesis-heavy questions. The paper isolates a vector-retrieval pipeline and does not use a structured graph representation.

Keywords

Medical retrieval, DenseRet, RetGen, Clinical evidence

1. Introduction

Medical QA benefits from domain knowledge, but a model's internal memory may be incomplete or outdated. RetGen [1] provides a retrieve-then-generate interface, and CuraLM-70B [2] offers a domain-adapted generator. MedRetGen combines them while keeping the evidence in text passages only. The central question is whether DenseRet can supply useful clinical context without creating an additional reasoning layer.

2. Method

DenseRet encodes passages from medical textbooks and guidelines into a searchable vector index. At query time, the question vector retrieves the top passages, which are concatenated with the question and given to CuraLM-70B. The generator is trained to answer only after reading the context, with passage delimiters and source identifiers retained. A contradiction-aware preprocessing step flags passages that disagree on numbers or negation, but it does not resolve them. We evaluate the same generator with and without retrieval on MedQA and PubMedQA. The evaluation emphasizes accuracy and includes an error taxonomy for contradiction, missing evidence, and failure to combine multiple facts.

3. Experiments

MedRetGen reaches 67.9% on MedQA versus 63.8% for CuraLM-70B, a 4.1-point gain. On PubMedQA it reaches 73.4% versus 71.2%. Retrieval helps questions with a single supporting passage, but contradictory passages reduce accuracy on synthesis questions. The main limitation is that the generator cannot reliably reason across multiple retrieved facts.

System	MedQA accuracy	PubMedQA accuracy
CuraLM-70B	63.8%	71.2%
MedRetGen	67.9%	73.4%

4. Conclusion

MedRetGen confirms that DenseRet and RetGen can improve CuraLM-70B on Medical QA. The gains are useful but modest because text retrieval may return contradictory evidence and does not guarantee multi-fact reasoning. Structured evidence selection is a natural next direction.

References

[1] Vik, A. (2021). Retrieve-then-Generate: Augmenting Language Models with Retrieved Passages for Knowledge-Intensive Tasks. Conference on Natural Language Systems (CNLS).

[2] Brandt, S. (2023). CuraLM-70B: Adapting a Large Language Model to Clinical Text. Journal of Clinical AI (JCAI).